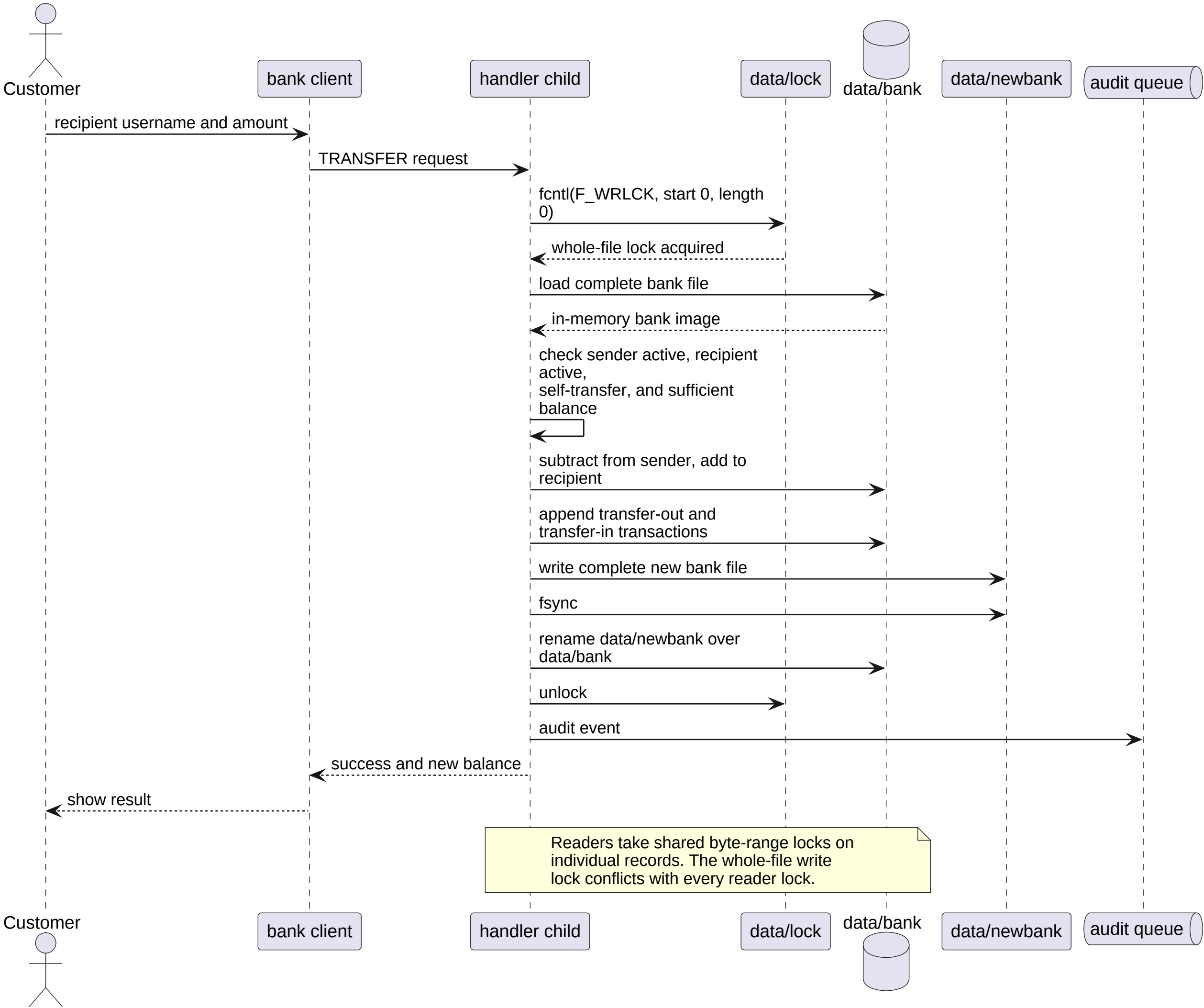
Customer
bank client
handler child
data/lock
data/bank
data/newbank
audit queue
recipient username and amount
TRANSFER request
fcntl(F_WRLCK, start 0, length 0)
whole-file lock acquired
load complete bank file
in-memory bank image
check sender active, recipient active,
self-transfer, and sufficient balance
subtract from sender, add to recipient
append transfer-out and transfer-in transactions
write complete new bank file
fsync
rename data/newbank over data/bank
unlock
audit event
success and new balance
show result
Readers take shared byte-range locks on individual records. The whole-file write lock conflicts with every reader lock.
Customer
bank client
handler child
data/lock
data/bank
data/newbank
audit queue